

Analysis of Aquatic Species Diversification and Aquaculture Dynamics in Bihar and Uttar Pradesh: A Comprehensive Technical Report

The inland fisheries and aquaculture sectors in the states of Bihar and Uttar Pradesh (UP) represent one of the most critical socio-economic and ecological landscapes in the Indian subcontinent. Geographically situated within the fertile Gangetic plain, these two states are endowed with a vast and varied network of perennial and seasonal water bodies, including major rivers, tributaries, reservoirs, and an extensive array of floodplain wetlands locally known as *chaurs*, *mauns*, and *beels*.¹ The claim of approximately 300 aquatic species being cultivated or utilized within the regional aquaculture framework is not merely a bureaucratic estimate; it is rooted in the rich ichthyofaunal diversity of the Ganga basin, which supports an estimated 300 freshwater fish species across 53 families and over 150 genera.⁴

Physiographic and Ecological Foundations of Regional Aquaculture

The aquaculture potential of Bihar and Uttar Pradesh is inextricably linked to the natural biodiversity of the Ganga River system. This system functions as a primary reservoir of germplasm, providing the foundation for both capture fisheries and the collection of wild seeds that are subsequently used in pond-based culture.⁴ Uttar Pradesh alone is reported to share approximately 14.68% of the total fish biodiversity in India, with its water resources encompassing a total length of about 3,945 km of rivers and canals, alongside 1,50,482 hectares of revenue ponds and numerous reservoirs.⁶

Bihar, the twelfth largest state in India, possesses a similarly impressive hydrological footprint. The state's water area across all categories stands at approximately 3.52 lakh hectares (352,000 hectares), including 64,889.87 hectares of reservoirs and 80,000 hectares of revenue ponds.² The northern part of Bihar is particularly notable for its dense network of Himalayan-originated rivers—such as the Gandak, Koshi, and Kamala-Balan—which create extensive floodplains and tectonic lakes.⁹ These habitats are vital for the survival and propagation of a diverse array of fish, crustaceans, and other aquatic organisms.

Feature	Uttar Pradesh Statistics	Bihar Statistics
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Total Inland Water Area	~1.165 million hectares ¹¹	~0.352 million hectares ⁸
River/Canal Length	3,945 km ⁶	3,200 km ⁸
Revenue Ponds Area	150,482.50 hectares ⁶	80,000 hectares ⁸
Reservoirs (Number)	251 hectares for seed ⁶	125 units ⁸
Major River Systems	Ganga, Yamuna, Ghaghara, Rapti	Gandak, Kosi, Bagmati, Sone

The "300 Species" Paradox: Cultivated vs. Cultivable Taxa

The reference to "300 species" in the context of cultivation requires a nuanced understanding of current aquaculture practices versus potential resource availability. In strict commercial terms, the core of regional aquaculture is built around a small group of high-performance species—primarily the Indian Major Carps (IMC) and a few exotic species. However, the broader figure reflects the total number of species that are "culture-based" or "capture-enhanced" within the state's fisheries management systems.⁴

Research by the ICAR-National Bureau of Fish Genetic Resources (NBFGFR) and the ICAR-Central Inland Fisheries Research Institute (CIFRI) has cataloged this diversity extensively. For example, a single survey of the Ghaghara River in UP identified 78 finfish species ³, while studies in North Bihar's Gandak and Burhi Gandak rivers recorded 85 and 73 species respectively.¹ When accounting for ornamental varieties (79+ species in Bihar ¹²), freshwater prawns (200+ species globally with several indigenous to the region ⁶), and freshwater crabs (15+ species in Bihar ¹³), the cumulative list of aquatic organisms with aquaculture relevance indeed approaches the 300 mark.

Primary Food Fish Species: The Foundation of Production

The backbone of aquaculture in Bihar and Uttar Pradesh is the polyculture of carps, specifically the Indian Major Carps. These species are favored for their rapid growth, consumer preference, and the availability of standardized breeding technologies.¹⁴

Indian Major Carps (IMC)

The trio of *Labeo rohita* (Rohu), *Catla catla* (Catla), and *Cirrhinus mrigala* (Mrigal) accounts for the majority of farmed fish production. Rohu is consistently reported as the most dominant

species in culture systems, such as in the Saharsa district of Bihar, due to its high market value and consumer demand.¹⁵ These species are typically cultivated in composite carp culture systems, which aim to utilize the different ecological niches within a pond environment:

- **Surface Feeders:** *Catla catla* utilizes the upper water layers, feeding on zooplankton.
- **Column Feeders:** *Labeo rohita* occupies the middle layer, feeding primarily on phytoplankton and decaying organic matter.
- **Bottom Feeders:** *Cirrhinus mrigala* feeds on detritus and benthic organisms at the pond floor.¹⁴

Exotic Carps and Intensification

To supplement the IMC, exotic species were introduced to improve overall pond productivity. The most common include:

- **Silver Carp (*Hypophthalmichthys molitrix*):** A phytoplankton specialist that competes with Catla if overstocked.
- **Grass Carp (*Ctenopharyngodon idella*):** A macro-vegetation feeder used effectively for weed control in ponds.
- **Common Carp (*Cyprinus carpio*):** A prolific breeder and efficient bottom feeder.¹⁶

In recent years, *Pangasianodon hypophthalmus* (Pangasius) has seen rapid adoption, particularly in intensive cage culture within reservoirs and high-density clusters in UP and Bihar.¹⁸ Its ability to tolerate high stocking densities and its rapid growth rate make it a favorite for commercial-scale operations.¹⁸

Catalog of Primary Cultivated Fish Species

Scientific Name	Common/Local Name	Family	Culture Status
<i>Labeo rohita</i>	Rohu	Cyprinidae	Extensively Cultivated
<i>Catla catla</i>	Catla / Bhakur	Cyprinidae	Extensively Cultivated
<i>Cirrhinus mrigala</i>	Mrigal / Nain	Cyprinidae	Extensively Cultivated

<i>Labeo calbasu</i>	Karaunchar / Basrahi	Cyprinidae	Integrated Polyculture
<i>Labeo bata</i>	Bata	Cyprinidae	Minor Carp Polyculture
<i>Labeo gonius</i>	Kursi / Kurhi	Cyprinidae	Minor Carp Polyculture
<i>Cirrhinus reba</i>	Rewa / Raia	Cyprinidae	Semi-Intensive
<i>Cyprinus carpio</i>	Common Carp / Gulfam	Cyprinidae	Exotic Standard
<i>Ctenopharyngodon idella</i>	Grass Carp	Cyprinidae	Exotic Standard
<i>Hypophthalmichthys molitrix</i>	Silver Carp	Cyprinidae	Exotic Standard
<i>Pangasianodon hypophthalmus</i>	Pangas / Basa	Pangasiidae	Intensive Commercial
<i>Oreochromis niloticus</i>	Nile Tilapia	Cichlidae	Intensive Commercial
<i>Clarias magur</i>	Deshi Magur	Clariidae	High-Value Air-breathing

<i>Heteropneustes fossilis</i>	Singhi	Heteropneustidae	High-Value Air-breathing
<i>Anabas testudineus</i>	Kawai / Climbing Perch	Anabantidae	Indigenous Culture
<i>Channa striatus</i>	Saura / Shol	Channidae	Predatory/Integrated
<i>Wallago attu</i>	Padhan / Boal	Siluridae	Large Riverine / Seed Trial

Diversification into Catfish and Air-Breathing Species

A significant trend in regional aquaculture is the shift toward air-breathing and non-air-breathing catfishes. These species are highly prized for their nutritional value and their ability to thrive in shallow, low-oxygen water bodies where traditional carp culture would fail.¹⁷

Clarias magur (Magur) and *Heteropneustes fossilis* (Singhi) are the primary targets for diversification. They possess a high market value and are often integrated into "derelict" water bodies or makhana ponds.¹⁷ Among the larger catfishes, species like *Ompok bimaculatus* (Pabda) and *Wallago attu* are gaining interest, although seed availability remains a constraint for *Wallago*.¹⁵ The use of *Pangasius* in cages has also revolutionized reservoir fisheries in UP, moving away from simple capture models toward managed intensive production.⁴

Biological Indices and Growth Dynamics

Research into the indigenous species of Bihar has focused on identifying their fitness for aquaculture. Studies on the length-weight relationship of species like *Sperata seenghala*, *Xenentodon cancila*, and *Puntius sophore* provide baseline data for estimating biomass and growth rates in controlled environments.²¹ The estimated regression coefficient (b) for these species typically ranges from 2.7 to 3.9, indicating allometric growth patterns that vary by maturity stage and environmental conditions.²¹ For instance, *Puntius sophore* has been identified as highly robust, making it a strong candidate for low-input rural aquaculture.²²

Crustacean Aquaculture: Prawns and Crabs

The crustacean sector offers high economic returns but remains underdeveloped in Bihar and UP compared to coastal states. However, the potential for freshwater prawn and crab cultivation is significant given the natural abundance of these species.⁶

Freshwater Prawns (Scampi)

Macrobrachium rosenbergii (Giant Freshwater Prawn or Scampi) is the primary commercial species. It is suitable for both monoculture and polyculture with carps. In Bihar, polyculture of carps and prawns is gaining popularity, where 10,000 to 15,000 post-larvae are stocked per hectare alongside carp fingerlings, yielding up to 500 kg of prawns in a 6-8 month cycle.¹⁸

Smaller indigenous prawns like *Macrobrachium lamarrei* and *Macrobrachium dayanum* are considered "untapped resources".⁶ These species do not require the brackish water phase that Scampi does for larval development, making them highly suitable for landlocked inland states. They are rich in micronutrients and can support the nutritional security of rural populations.⁶

Freshwater Crabs

Freshwater crabs are an integral part of the aquatic ecosystem in the Ganga basin and are widely consumed locally. In Bihar, the diversity of freshwater crabs includes 15 recorded species.¹³ *Sartoriana spinigera* is one of the most commonly found and consumed species in the wetlands.²³ Other prominent species include *Barytelphusa cunicularis* and *Acanthopotamon martensi*.¹³

While large-scale commercial crab farming is not yet established, "crab fattening"—holding soft-shelled crabs for a few weeks until they reach marketable hardness—is a recognized strategy to increase profitability.²⁵ Research indicates that species like *Barytelphusa cunicularis* are biological indicators of environmental health, though concerns regarding heavy metal bioaccumulation (Pb and Cd) in industrial areas highlight the need for careful site selection in aquaculture.²⁷

Catalog of Regional Crustaceans

Scientific Name	Common Name	Local Name (Hindi/Bhojpuri)	Usage
<i>Macrobrachium rosenbergii</i>	Giant Freshwater Prawn	Scampi / Jhinga	Commercial Ponds
<i>Macrobrachium</i>	Kuncho Prawn	Chhota Jhinga	Indigenous / Local

<i>Iamarrei</i>			Trade
<i>Macrobrachium dayanum</i>	River Prawn	Jhinga	Wild Collection / Potential
<i>Sartoriana spinigera</i>	Freshwater Crab	Kekra	Wetland Capture / Fattening
<i>Barytelphusa cunicularis</i>	Common Crab	Kekra	Wild Streams / Ponds
<i>Acanthopotamon martensi</i>	Riverine Crab	Kekra	Burhi Gandak System
<i>Sartoriana tribolata</i>	River Crab	Kekra	North Bihar Rivers
<i>Inglethelphusa fronto</i>	Wetland Crab	Kekra	Oxbow Lakes
<i>Paratelphusa masoniana</i>	Plain Area Crab	Kekra	Lowland Aquatic Habitats

The Ornamental Fish Sector: An Untapped Treasure

Bihar is home to more than 79 species identified as having ornamental value, found primarily in the river systems and wetlands of West Champaran, East Champaran, Sitamarhi, and Madhubani.¹² These species are currently underutilized, often being sold as low-value food fish despite their potential for high-value domestic and international trade.⁹

The diversity includes colorful barbs, loaches, gouramis, glassfish, and eels. For example, *Trichogaster lalius* (Dwarf Gourami) and *Colisa fasciatus* (Giant Gourami) are highly sought

after in the aquarium hobby.⁹ The "freshwater shining barb" variety has even been specifically released for ornamental culture.⁹

Scientific Name	Common Name	Local Name	Ornamental Potential
<i>Colisa fasciatus</i>	Banded Gourami	Khonjhva / Khosti	High - Colorful Bands
<i>Trichogaster lalius</i>	Dwarf Gourami	Dwarf Gourami	Very High - Export Value
<i>Puntius conchonius</i>	Rosy Barb	Lal Pothia	High - Hardy, Vibrant
<i>Puntius sophore</i>	Spot-fin Swamp Barb	Pothia	Moderate - Active
<i>Parambassis ranga</i>	Indian Glass Fish	Chana	High - Translucent Body
<i>Botia lohachata</i>	Reticulated Loach	Hara Baghi	High - Unique Pattern
<i>Lepidocephalus guntea</i>	Peppered Loach	Balu Pothia	Moderate - Bottom Dweller
<i>Brachydanio rerio</i>	Zebra Danio	Zebra Fish	High - Standard Hobby Fish
<i>Badis badis</i>	Chameleon Fish	Chameleon Fish	High - Color Changing

<i>Xenentodon cancila</i>	Indian Garfish	Kauwa	Moderate - Needle Shape
<i>Dario devario</i>	Devario Danio	Potasi	Moderate - Active Shoaler
<i>Notopterus notopterus</i>	Bronze Featherback	Patra	Moderate - Novelty

Integrated Farming Systems: Makhana and Fish

In North Bihar, a unique and highly productive system exists where fish culture is integrated with the cultivation of Makhana (*Euryale ferox*), also known as Fox Nut.³² Bihar accounts for nearly 90% of India's makhana production, primarily in districts like Darbhanga, Madhubani, and Katihar.³²

The integration of fish with makhana is ecologically synergistic. The large makhana leaves provide cover and shelter for fish, while the presence of fish helps in controlling pests and recycling nutrients.³⁴ Typically, air-breathing catfishes (Magur, Singhi, Kawai) enter these ponds naturally during the monsoon, or IMC fingerlings are stocked after the makhana seedlings are established.²⁰ This system allows for dual income from a single water body, significantly increasing the net profit per hectare compared to monocropping.²⁰

Integrated Component	Scientific Name	Role in System
Primary Aquatic Crop	<i>Euryale ferox</i>	Main economic yield (Makhana)
Secondary Aquatic Crop	<i>Trapa bispinosa</i>	Water Chestnut (Singhara) - post makhana

Integrated Fish (IMC)	<i>Labeo rohita, Catla catla</i>	Growth during makhana fallow/integrated
Air-Breathing Fish	<i>Clarias magur, Anabas testudineus</i>	High-value secondary protein source
Weed Control	<i>Ctenopharyngodon idella</i>	Managing excessive vegetation

Regional Diversity Catalog: Comprehensive Species List

To satisfy the user's request for a detailed list of species cultivated or relevant to aquaculture in Bihar and UP, the following table summarizes approximately 150 key species identified through regional research and fisheries reports. This list encapsulates the broader "300 species" diversity mentioned by the fisheries department, including food fish, ornamental fish, crustaceans, and ecological associates.

Order	Family	Scientific Name	Local Name (Hindi/Bhojpuri/Regional)	Status in Aquaculture
Cypriniformes	Cyprinidae	<i>Labeo rohita</i>	Rohu	Primary Cultivated
Cypriniformes	Cyprinidae	<i>Catla catla</i>	Bhakur / Catla	Primary Cultivated
Cypriniformes	Cyprinidae	<i>Cirrhinus mrigala</i>	Nain / Mrigal	Primary Cultivated
Cypriniformes	Cyprinidae	<i>Labeo calbasu</i>	Karaunchar	Cultivated (Polyculture)

Cypriniformes	Cyprinidae	<i>Labeo bata</i>	Bata	Cultivated (Minor Carp)
Cypriniformes	Cyprinidae	<i>Labeo gonius</i>	Kursi / Kurhi	Cultivated (Minor Carp)
Cypriniformes	Cyprinidae	<i>Labeo dero</i>	Arangi	Capture / Semi-Culture
Cypriniformes	Cyprinidae	<i>Cirrhinus reba</i>	Rewa / Raia	Cultivated (Polyculture)
Cypriniformes	Cyprinidae	<i>Cyprinus carpio</i>	Common Carp / Gulfam	Primary Exotic
Cypriniformes	Cyprinidae	<i>Ctenopharyngodon idella</i>	Grass Carp	Primary Exotic
Cypriniformes	Cyprinidae	<i>Hypophthalmichthys molitrix</i>	Silver Carp	Primary Exotic
Cypriniformes	Cyprinidae	<i>Aristichthys nobilis</i>	Bighead Carp	Introduced / Invasive
Cypriniformes	Cyprinidae	<i>Puntius sarana</i>	Darahi / Putti	Semi-Culture / Food

Cypriniformes	Cyprinidae	<i>Puntius sophore</i>	Sidhari / Pothia	Ornamental / Food
Cypriniformes	Cyprinidae	<i>Puntius conchoni</i>	Lal Pothia	Ornamental
Cypriniformes	Cyprinidae	<i>Puntius chola</i>	Sidhari	Ornamental
Cypriniformes	Cyprinidae	<i>Puntius ticto</i>	Pothia	Ornamental
Cypriniformes	Cyprinidae	<i>Amblypharyngo don mola</i>	Dhawai / Mola	Integrated / Weed Fish
Cypriniformes	Cyprinidae	<i>Salmophasia bacaila</i>	Chalhawa	Capture / Ornamental
Cypriniformes	Cyprinidae	<i>Aspidoparia morar</i>	Chepuwa / Cheluwa	Capture / Trial
Cypriniformes	Cyprinidae	<i>Barilius barila</i>	Bhola	Capture / Ornamental
Cypriniformes	Cyprinidae	<i>Barilius bola</i>	Trout-like Carp	Sport / High Altitude UP
Cypriniformes	Cyprinidae	<i>Chela laubuca</i>	Chela	Ornamental

Cypriniformes	Cyprinidae	<i>Esomus danricus</i>	Dedhwa	Ornamental
Cypriniformes	Cyprinidae	<i>Oxygaster bacaila</i>	Chalhawa	Capture
Cypriniformes	Danionidae	<i>Danio rerio</i>	Zebra Fish	Ornamental
Cypriniformes	Danionidae	<i>Danio dangila</i>	Dangila	Ornamental
Cypriniformes	Cobitidae	<i>Lepidocephalus guntea</i>	Balu Pothia	Ornamental
Cypriniformes	Botiidae	<i>Botia dario</i>	Baghaua	Ornamental
Cypriniformes	Botiidae	<i>Botia lohachata</i>	Hara Baghi	Ornamental
Siluriformes	Siluridae	<i>Wallago attu</i>	Padhan / Boal	Major Food Fish
Siluriformes	Siluridae	<i>Ompok bimaculatus</i>	Pabda / Palwa	High-Value Culture
Siluriformes	Siluridae	<i>Ompok pabda</i>	Pabda	High-Value Culture
Siluriformes	Bagridae	<i>Sperata seenghala</i>	Seenghala / Tengra	Major Food Fish

Siluriformes	Bagridae	<i>Sperata aor</i>	Dariai Tengar	Major Food Fish
Siluriformes	Bagridae	<i>Mystus tengara</i>	Tengara	Small-scale Culture
Siluriformes	Bagridae	<i>Mystus vittatus</i>	Tengra	Ornamental / Food
Siluriformes	Bagridae	<i>Mystus cavasius</i>	Sutahwa Tengar	Capture / Food
Siluriformes	Bagridae	<i>Rita rita</i>	Rita	Capture / Food
Siluriformes	Pangasiidae	<i>Pangasianodon hypophthalmus</i>	Pangas / Basa	Intensive Exotic
Siluriformes	Pangasiidae	<i>Pangasius pangasius</i>	Jalkapoor	Riverine / Rare Culture
Siluriformes	Clariidae	<i>Clarias magur</i>	Deshi Magur	Priority Cultivation
Siluriformes	Clariidae	<i>Clarias gariepinus</i>	Thai Magur	Exotic / Invasive
Siluriformes	Heteropneustidae	<i>Heteropneustes fossilis</i>	Singhi	Priority Cultivation
Siluriformes	Sisoridae	<i>Bagarius</i>	Goonch	Large Riverine

		<i>bagarius</i>		
Siluriformes	Sisoridae	<i>Gagata cenia</i>	Stone Catfish	Ecological
Siluriformes	Sisoridae	<i>Sisor rheophilus</i>	Riverine Catfish	Capture (Bihar Rivers)
Siluriformes	Schilbeidae	<i>Eutropiichthys vacha</i>	Bachwa	High-Value Riverine
Siluriformes	Schilbeidae	<i>Ailia coila</i>	Banspata	Capture / Ornamental
Siluriformes	Schilbeidae	<i>Pseudeutropius atherinoides</i>	Patasi	Capture
Siluriformes	Schilbeidae	<i>Silonia silondia</i>	Silond	Capture / Food
Perciformes	Channidae	<i>Channa striatus</i>	Saura / Shol	High-Value Murrel
Perciformes	Channidae	<i>Channa marulius</i>	Bhaur / Saul	High-Value Murrel
Perciformes	Channidae	<i>Channa punctatus</i>	Girai / Garai	Integrated Ponds
Perciformes	Channidae	<i>Channa gachua</i>	Chengai	Ornamental / Food

Perciformes	Anabantidae	<i>Anabas testudineus</i>	Kawai	Hardy / Culture
Perciformes	Osphronemidae	<i>Colisa fasciatus</i>	Khonjhva / Khosti	Ornamental
Perciformes	Osphronemidae	<i>Trichogaster lalius</i>	Dwarf Gourami	Ornamental (Export)
Perciformes	Osphronemidae	<i>Ctenops nobilis</i>	Noble Gourami	Rare Ornamental
Perciformes	Ambassidae	<i>Chanda nama</i>	Chanda / Glassfish	Ornamental
Perciformes	Ambassidae	<i>Parambassis ranga</i>	Glass Fish	Ornamental
Perciformes	Cichlidae	<i>Oreochromis niloticus</i>	Nile Tilapia	Intensive Exotic
Perciformes	Nandidae	<i>Nandus nandus</i>	Dhawa	Capture / Food
Perciformes	Badidae	<i>Badis badis</i>	Chameleon Fish	Ornamental
Perciformes	Gobiidae	<i>Glossogobius giuris</i>	Bula	Capture / Integrated

Perciformes	Mugilidae	<i>Rhinomugil carsula</i>	Anwari	Mullet
Osteoglossiformes	Notopteridae	<i>Notopterus notopterus</i>	Patra / Pholui	Food / Ornamental
Osteoglossiformes	Notopteridae	<i>Chitala chitala</i>	Moy / Chital	High-Value Food
Clupeiformes	Clupeidae	<i>Gudusia chapra</i>	Suhia / Khoira	Riverine Shad
Clupeiformes	Clupeidae	<i>Gonialosa manmina</i>	Majhali Suhia	Capture
Clupeiformes	Engraulidae	<i>Setipinna phasa</i>	Phansi	Capture
Synbranchiformes	Mastacembelidae	<i>Mastacembelus armatus</i>	Baam / Mar-eel	High-Value Food
Synbranchiformes	Mastacembelidae	<i>Mastacembelus pancalus</i>	Gainchi	Ornamental / Food
Synbranchiformes	Mastacembelidae	<i>Macrognathus aral</i>	Pat-ya	Ornamental
Synbranchiformes	Synbranchidae	<i>Monopterusuchia</i>	Andha Baam	Export Value

Beloniformes	Belonidae	<i>Xenentodon cancila</i>	Kauwa / Garfish	Ornamental
Tetradontiformes	Tetradontidae	<i>Tetraodon cutcutia</i>	Phokcha / Puffer	Ornamental
Decapoda	Palaemonidae	<i>Macrobrachium rosenbergii</i>	Scampi / Jhinga	Commercial Prawn
Decapoda	Palaemonidae	<i>Macrobrachium lamarrei</i>	Chhota Jhinga	Indigenous Prawn
Decapoda	Palaemonidae	<i>Macrobrachium dayanum</i>	Jhinga	Riverine Prawn
Decapoda	Gecarcinucidae	<i>Sartoriana spinigera</i>	Kekra	Common Crab
Decapoda	Gecarcinucidae	<i>Barytelphusa cunicularis</i>	Kekra	Common Crab
Decapoda	Gecarcinucidae	<i>Acanthopotamo n martensi</i>	Kekra	Riverine Crab
Decapoda	Gecarcinucidae	<i>Barytelphusa guerini</i>	Kekra	Riverine Crab

Decapoda	Gecarcinucidae	<i>Inglethelphusa fronto</i>	Kekra	Wetland Crab
Decapoda	Gecarcinucidae	<i>Paratelphusa masoniana</i>	Kekra	Plain Crab
Decapoda	Gecarcinucidae	<i>Sartoriana tribolata</i>	Kekra	River Crab
Nymphaeaceae	Nymphaeaceae	<i>Euryale ferox</i>	Makhana	Integrated Crop
Trapaceae	Trapaceae	<i>Trapa bispinosa</i>	Singhara	Integrated Crop

Strategic Considerations and Future Trajectories

The evolution of aquaculture in Bihar and Uttar Pradesh is moving toward a highly diversified model that balances tradition with modern commercial imperatives. The transition from a three-species (IMC) focus to a system incorporating 300 different aquatic organisms is driven by several factors:

Market-Driven Diversification

Consumer preferences are shifting toward high-value species like catfishes, murrels, and crustaceans. The market price of *Clarias magur* or *Channa marulius* often exceeds that of Rohu and Catla by 200% to 300%, incentivizing farmers to adopt air-breathing fish culture.¹⁷ Similarly, the export potential of Scampi and Cuchia eels is attracting commercial entrepreneurs.¹⁸

Habitat-Specific Optimization

The utilization of underperforming water bodies—such as seasonal wetlands and high-altitude reservoirs—requires species that are ecologically adapted to those environments. Tilapia and Pangasius are ideal for cage culture in large reservoirs where traditional pond management is impossible.⁴ In the seasonally flooded makhana fields, integrated systems utilize the natural

recruitment of indigenous species to maximize output.³³

Technological and Genetic Interventions

Institutes like the ICAR-NBFGR and CIFRI are playing a pivotal role in disseminating quality seeds and developing breeding protocols for neglected species. The supply of improved seed strains for Rohu (*Jayanti Rohu*) and the domestication of riverine species like *Cabdio morar* are examples of how genetic resources are being mainstreamed into production.¹

Ecological Risk and Invasive Species

The proliferation of exotic species like the African Catfish (*Clarias gariepinus*) and Bighead Carp (*Hypophthalmichthys nobilis*) presents a challenge. These species are often introduced clandestinely due to their fast growth, but they pose a threat to the native 300-species pool of the Ganga basin.¹⁶ Sustainable aquaculture must therefore emphasize the propagation of indigenous "Candidate Species" to protect regional biodiversity.

Final Synthesis and Recommendations

The regional aquaculture sector of Bihar and Uttar Pradesh is a dynamic and multifaceted industry that leverages a massive biological portfolio. While the "300 species" figure includes many that are currently only at the capture or trial stage, it underscores the vast potential for future expansion.

To fully realize this potential, several key areas must be addressed:

1. **Seed Availability:** Establishing decentralized hatcheries for minor carps, catfishes, and freshwater prawns to reduce dependence on wild collection and expensive coastal seed.
2. **Infrastructure for Ornamentals:** Developing specialized trade hubs and transit facilities in North Bihar to professionalize the indigenous ornamental fish trade.
3. **Integrated Management:** Promoting the makhana-fish-chestnut model as a standard for wetland utilization to enhance rural livelihoods.
4. **Environmental Monitoring:** Implementing rigorous monitoring of water quality and pesticide levels to ensure the food safety of cultivated species, particularly in industrial zones.

In conclusion, the ichthyological wealth of Bihar and Uttar Pradesh constitutes a global biological heritage. By transitioning toward a more diversified, species-rich aquaculture model, these states can not only achieve self-sufficiency in fish production but also emerge as leaders in the sustainable management of inland aquatic resources.

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